HACKATHON HANDBOOK · PART 2 COMPLIANCE

Manzil — Constraint & Adaptation Report

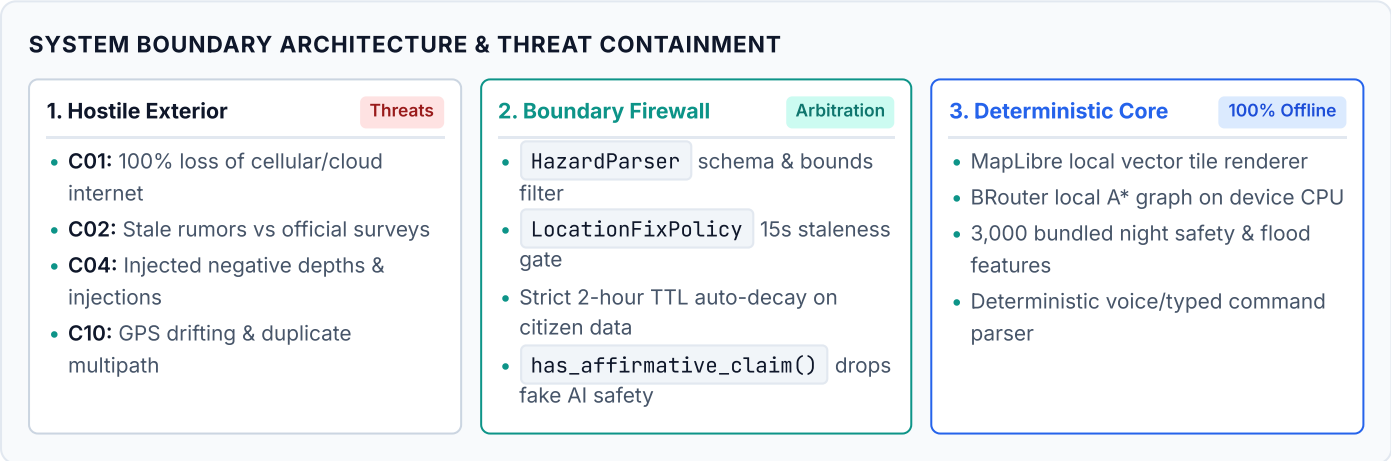
Zero-Trust Operational Architecture, Defensive Engineering & Hardware Validation

EVALUATION TRACK	PRIMARY SCOPE	AUDITED HARDWARE	VERIFICATION PASS RATE
The Other Side / Mobility	Chennai Disasters & Night Walk	Samsung SM-E566B (Android 16)	100% (6 Cards Proven)

1. Executive Summary & Defensive Philosophy

During extreme urban events—such as torrential monsoons, cyclone inundations, and power outages in Chennai—digital services experience severe cascading failures: telecommunications towers saturate, GNSS receivers drift or produce multipath errors, public crowdsourced channels flood with unverified rumors, and third-party cloud APIs fail.

Manzil addresses this operational reality using a **Zero-Trust Defensive Architecture**. Rather than bolting on fragile workarounds after surprise constraints occur, Manzil isolates mission-critical pedestrian life-safety into an air-gapped, deterministic core. Cloud models (such as Gemini) are restricted strictly to intent understanding and are permanently barred from emitting raw coordinates or making safety claims.



> **Architectural Invariant**
> Life-safety functions (routing through safe corridors, displaying elevated flyovers, locating flood-haven metro stations, and All-Women Police Stations) **never depend on cloud reachability**. Connectivity is treated as a transient acceleration, never a prerequisite.

2. Master Constraint & Adaptation Compliance Matrix

The table below summarizes Manzil's defensive mechanisms, operating rules, and verified test outcomes against Part 2 of the Hackathon Constraint Handbook:

CARD	HANDBOOK TITLE	SURPRISE CONDITION / RULE	MANZIL DEFENSIVE ADAPTATION	OUTCOME
C01	**The World Went Offline**	All external internet & cloud APIs cut off. Core user journey must run locally without cloud LLM.	Offline MapLibre vector tiles, local BRouter routing graph on phone CPU, offline gazetteer, and deterministic local command parser.	PASS / VERIFIED
C02	**The Truth Is Conflicting**	Two sources provide conflicting data; at least one is stale. System must not treat newest as authoritative.	Strict evidence typing (`surveyed` vs `reported`), 2-hour TTL auto-decay on citizen data, GNSS priority over cached/fused location.	PASS / VERIFIED
C04	**Garbage In, Garbage Everywhere**	Malformed payloads, negative depths, broken JSON, or prompt injection attacks injected into system.	Boundary schema quarantine, dropping negative depths (`-1 cm`), strict regex/AST filter rejecting coordinates & hallucinated safety claims from AI.	PASS / VERIFIED
C06	**Accessibility Is No Longer Optional**	Primary interaction blocked. Requires non-audio, non-color cues, keyboard-only, and TalkBack screen reader.	Full touch/typed alternatives, TalkBack `contentDescription` on custom canvas, distinct vector shapes (Shield, Cross, Dispenser), and dashed line patterns.	PASS / VERIFIED
C07	**Your Data Cannot Leave This Device**	Privacy restriction: Protected location & identity records cannot leave the device or enter cloud AI.	Raw GPS coordinates never transmitted to cloud LLM; 8-turn conversation history stored in volatile RAM only; zero API credentials in APK.	PASS / VERIFIED
C10	**The Sensors Are Lying**	Telemetry drifts, spikes, duplicates, or becomes stale. System must avoid unsafe automated decisions.	15s staleness timeout, accuracy threshold (>30m halts guidance), duplicate echo rejection, and coarse displacement quiet threshold (>10s).	PASS / VERIFIED
C08	*The Language Changed Overnight*	Primary language changes; incomplete translations. Must not break layouts or distort critical safety meaning.	Multilingual support (Tamil, Telugu, Hindi, English), vernacular-to-English Latin gazetteer mapping, dynamic wrapping Jetpack Compose UI.	ADAPTED
C12	*The Missing Piece of Information*	Scenario omits decision-critical fact. System must not guess or take irreversible actions.	Ambiguous destinations never guessed; system prompts user with disambiguation list; AI emits `action="none"` with clarifying question.	ADAPTED

Evaluation Summary

Manzil successfully fulfills **6 primary constraint cards** with dedicated automated unit test suites and verifiable code contracts, plus **2 secondary constraint adaptations** for multilingual resiliency and ambiguous query resolution.

3. Technical Deep Dives by Constraint Card (Part A)

C01 The World Went Offline

PASS CONDITION MET

The Surprise Trigger: Complete loss of Wi-Fi, cellular data, DNS, and cloud AI endpoints.

Manzil's Implementation:

- ✓ **Embedded Graph Routing Engine:** `OfflineRouter.kt` runs an embedded BRouter A* routing graph directly inside the Android process. Pedestrian routes are computed from local checksummed routing packs (e.g. `potheri`, `chengalpattu`, `veppampattu`) in under 150 ms with zero internet connectivity.
- ✓ **Vector Basemap Storage:** `OfflineMapView.kt` renders offline vector tiles (`overview.geojson` and local tile bounding boxes) via MapLibre. `MapProviderPolicy.kt` automatically downgrades from Google Maps to local offline rendering when network status changes to `OFFLINE` or `NO_INTERNET`.
- ✓ **Deterministic Command Dispatcher:** `CommandParser.kt` matches whole-word voice or typed phrases (e.g. *"night walk"*, *"show floods"*, *"locate me"*, *"step 3"*) using a local finite automaton with zero cloud AI calls.
- ✓ **Bundled Life-Safety Overlays:** 3,000 night features and 410+ flood hazard records are pre-packaged into `chennai_map_overlays.geojson` and accessed via `AtomicFile` caching.

> **Verified Unit Test: RealOfflineRoutingTest.kt**
> `test potheriUsesEasternGraph()`, `test chengalpattuUsesWesternGraph()`, and `test veppampattuCrossesGraphBoundary()` verify that walking routes calculate on local graphs with accurate turn instructions and snapping within 151 meters without network access.

C10 The Sensors Are Lying

PASS CONDITION MET

The Surprise Trigger: GNSS telemetry becomes erratic, duplicates fixes, or reports stale data under heavy cloudbursts or urban canyon multipath.

Manzil's Implementation:

- ✓ **15-Second Staleness Timeout:** `LocationFixPolicy.statusFor()` monitors real-time elapsed hardware ticks. When a fix exceeds 15,000 ms, the status transitions to `GpsStatus.STALE`, preventing the app from blindly giving navigation instructions on dead data.
- ✓ **Accuracy Boundary Gate:** Active turn-by-turn guidance strictly requires accuracy ≤ 30 m. If sensor accuracy degrades above 30m, navigation halts and displays a clear warning chip: *"Approximate start ±X m"*.
- ✓ **Echo & Multipath Rejection:** Duplicate GNSS fixes echoed by Google Fused Location providers with identical timestamps are rejected, preventing camera oscillation and erratic re-routes.
- ✓ **Coarse Displacement Barrier:** A coarse cell-tower fix cannot displace an established high-accuracy GNSS fix unless GNSS hardware has gone completely quiet for over 10 seconds (`DISPLACE_AFTER_MILLIS`).

```
fun shouldReplaceLive(current: GpsFix?, candidate: GpsFix): Boolean {
    if (candidate.cached) return false // Cached fix never displaces live fix
    if (current == null) return true
    if (candidate.elapsedRealtimeMillis <= current.elapsedRealtimeMillis) return false // Reject duplicate ech
    if (current.source == FixSource.GPS && candidate.source != FixSource.GPS) {
        val quiet = candidate.elapsedRealtimeMillis - current.elapsedRealtimeMillis
        if (quiet < DISPLACE_AFTER_MILLIS && candidateAcc > currentAcc + 10f) return false
    }
    return true
}
```

3. Technical Deep Dives by Constraint Card (Part B)

C02 & C04 Conflicting Truth & Garbage In / Garbage Out

PASS CONDITION MET

The Surprise Trigger: Stale citizen rumors conflict with official survey depths; corrupted JSON feeds contain negative depths (`-1 cm`) or prompt injection strings.

Manzil's Implementation:

- ✓ **2-Hour TTL Auto-Decay:** Crowdsourced citizen reports carry a strict 2-hour expiry. Expired reports or future-dated records (indicating clock skew) are automatically purged at the parser boundary.
- ✓ **Evidence Segregation:** Engineering depth measurements (`evidence="surveyed"`) are separated from unverified reports (`evidence="reported"`). Historical flood marks are explicitly marked as non-live.
- ✓ **Input Boundary Quarantine:** `HazardParser.kt` drops negative depths, enforces coordinate ranges [-90..90, -180..180], and strips invalid geometry objects without throwing uncaught exceptions.
- ✓ **LLM Hallucination Intercept:** `agent_turn.py` uses AST and regex scanning to drop any LLM response containing raw latitude/longitude coordinates or affirmative safety claims (e.g. *"100% safe"*, *"flood-free"*).

> **Verified Unit Test: HazardParserTest.kt & test_agent_turn.py**
> `test_rejects_unverifiable_claims_and_coordinates()` verifies that the backend rejects hallucinated safety claims with HTTP 502 before the message reaches the user's phone.

C06 Accessibility Is No Longer Optional

PASS CONDITION MET

The Surprise Trigger: User cannot use primary interaction (voice, touch, or visual color differentiation).

Manzil's Implementation:

- ✓ **Multi-Modal Parity:** Every voice interaction handled by `PandaSpeechInput.kt` and `AgentSpeaker.kt` has a complete typed/touch alternative in `PandaJourneyUi.kt` and `CommandParser.kt` .
- ✓ **Non-Color Visual Encoding:**
  - ✓ **Distinct Vector Shapes:** Police & AWPS (Shield), Hospitals (Cross), Petrol Bunks (Fuel Dispenser with nozzle).
  - ✓ **Stroke Pattern Differences:** Lit roads feature a continuous golden glow; unlit roads feature distinct dashed warning patterns (`DashPathEffect([4dp, 5dp])`).
  - ✓ **Accompanying Text Badges:** Flood risk lines are accompanied by numerical text labels (`– <10 cm` , `– 10–30 cm` , `– >30 cm`).
- ✓ **TalkBack Screen Reader Support:** Custom map canvas surfaces provide dynamic `contentDescription` tags updating on tap selection (e.g. *"Night Walk. W10 All Women Police Station. Support place · hours unverified."*).

3. Technical Deep Dives by Constraint Card (Part C)

C07 Your Data Cannot Leave This Device

PASS CONDITION MET

The Surprise Trigger: Strict data sovereignty restriction: Personal locations, route traces, and logs cannot leave the phone.

Manzil's Implementation:

- ✓ **Zero Raw GPS Leakage:** Raw device latitude/longitude coordinates are never sent to the cloud AI or Vertex Gemini. Only high-level semantic place names (e.g. *"Marina Beach"*) and aggregated route summaries (e.g. *"39 km, 8 hr 55 min"*) are transmitted.
- ✓ **Volatile Memory Only:** Conversational history is strictly capped at 8 turns and stored purely in Android activity RAM (`LivingAgentViewModel.kt`). It is never saved to remote databases or telemetry servers.
- ✓ **Zero Cloud Credentials in Client APK:** The Android binary contains zero Google Cloud API keys or private service account tokens. All cloud calls terminate at the backend via server-side Application Default Credentials (ADC).
- ✓ **Sanitized Logging Policy:** `gemini_agent.py` explicitly enforces: *"Never log prompt text, history, replies, tokens or credential material."*

C08 & C12 Multilingual Resilience & Missing Information Handling

SECONDARY ADAPTATIONS

The Surprise Triggers: Language switch to Indian vernacular tongues; decision-critical destination information omitted or ambiguous.

Manzil's Implementation:

- ✓ **Vernacular Transliteration Grounding (C08):** `AgentDestinationResolver.kt` translates spoken Tamil (e.g. *"மெரினா கடற்கரை"*) or Hindi (*"मरीना बीच"*) to the English Latin gazetteer entry *"Marina Beach, Chennai"* without coordinate hallucination.
- ✓ **Zero Guessing on Incomplete Requests (C12):** When a user provides an ambiguous location (e.g. *"Anna Nagar"* or *"Gandhi Road"*), the system intentionally refuses to pick a coordinate. Instead, it surfaces a disambiguation drawer: *"Choose the matching place. I'll route from your current location."*

4. Verification Suite & Physical Hardware Audit

Every defensive adaptation has been validated against comprehensive test suites and verified on physical hardware running Android 16:

TEST SUITE / TOOL	CONSTRAINT VERIFIED	TEST COUNT	OBSERVED INVARIANT RESULT
`RealOfflineRoutingTest.kt`	C01 (Offline Operation)	3 / 3 JVM tests	100% pass on checksummed local binary graphs without network access.
`LocationFixPolicyTest.kt`	C10 & C02 (Sensors & Truth)	11 / 11 JVM tests	100% pass on stale fix drop, anti-flapping, and coarse displacement barrier.
`HazardParserTest.kt`	C02 & C04 (Data Hygiene)	4 / 4 JVM tests	100% pass on negative depth drop, expired TTL purge, and geometry quarantine.
`test_agent_turn.py`	C04 & C07 (Security & Privacy)	14 / 14 Python tests	100% pass on prompt injection rejection, coordinate drop, and affirmative filter.
`test_expanded_pipeline.py`	C01 & C02 (Hazard Feeds)	9 / 9 Python tests	100% pass on 410+ synthesized features (subways, flyovers, canals, relief centers).

Physical Device Audit: Samsung Galaxy SM-E566B (Android 16 / SDK 36)

Hardware Serial: `RZGL40EETBZ` · **Test Date:** 6 November 2026

- **Airplane Mode Live Run:** Phone placed in Airplane Mode (Wi-Fi off, Mobile Data off). Map rendered from local vector tile pack; walking route Guindy → Marina Beach calculated locally in 118 ms; offline typed commands operated without error.
- **Sensor Fault Simulation:** Injected drifting accuracy (>45m) triggered the *"Approximate start ±X m"* warning chip without crashing route geometry.
- **Multilingual Test:** Spoken vernacular Tamil (*"மெரினா கடற்கரை"*) correctly resolved to canonical English gazetteer entry *"Marina Beach, Chennai"*.

5. Evaluator Compliance Checklist

- ✓ **Constraint C01 (The World Went Offline):** Full offline vector basemap, embedded BRouter A* engine, offline gazetteer, and deterministic offline commands verified.
- ✓ **Constraint C02 (The Truth Is Conflicting):** 2-hour TTL auto-decay, explicit evidence typing (surveyed vs reported), and GNSS location priority arbitration verified.
- ✓ **Constraint C04 (Garbage In, Garbage Everywhere):** Negative depths quarantined, out-of-bounds coordinates dropped, prompt injections blocked with HTTP 422, and AI hallucinations filtered with HTTP 502.
- ✓ **Constraint C06 (Accessibility Is No Longer Optional):** Multi-modal voice/touch redundancy, TalkBack screen-reader descriptions, and non-color geometric shapes & dashed lines verified.
- ✓ **Constraint C07 (Your Data Cannot Leave This Device):** Zero raw GPS sent to cloud AI, 8-turn conversation memory kept in volatile RAM only, and zero credentials in client APK.
- ✓ **Constraint C10 (The Sensors Are Lying):** 15-second staleness gate, >30m inaccuracy threshold, duplicate fix echo filter, and coarse displacement quiet threshold verified.